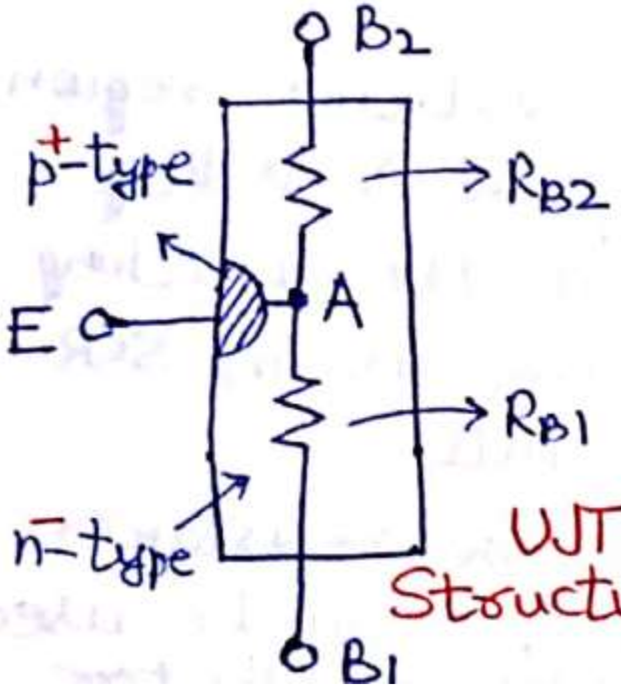
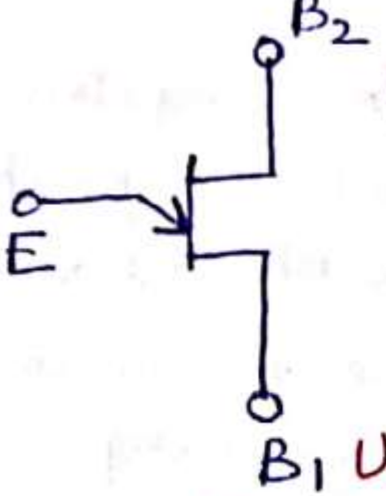
UJT

UNIJUNCTION TRANSISTOR (UJT)

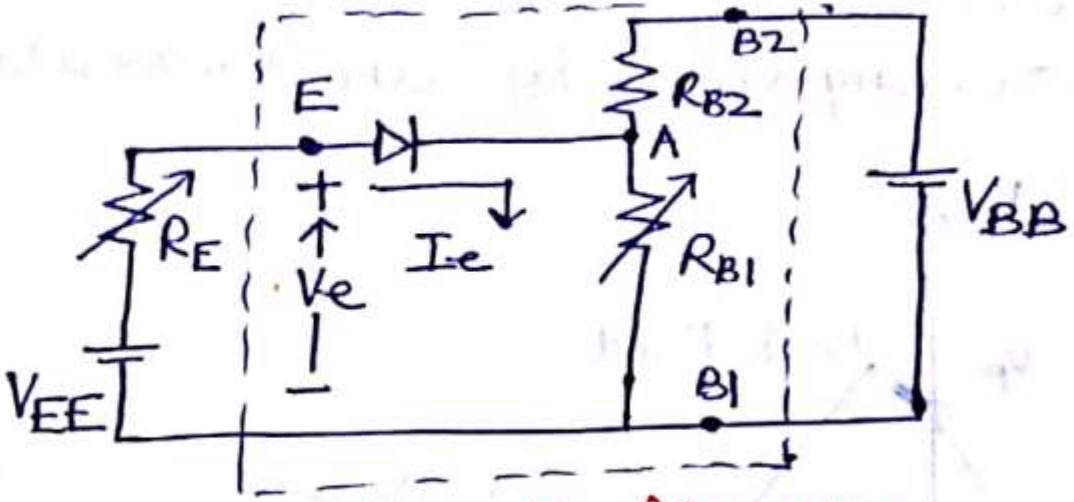
A UJT is made up of an n-type silicon base to which a p-type emitter is embedded. n-type base is lightly doped whereas p-type is heavily doped. The two ohmic contacts are called base 1 & base 2. Between B1 & B2, UJT behaves as an ordinary resistance.



UJT Structure



UJT Symbol



Circuit Diagram

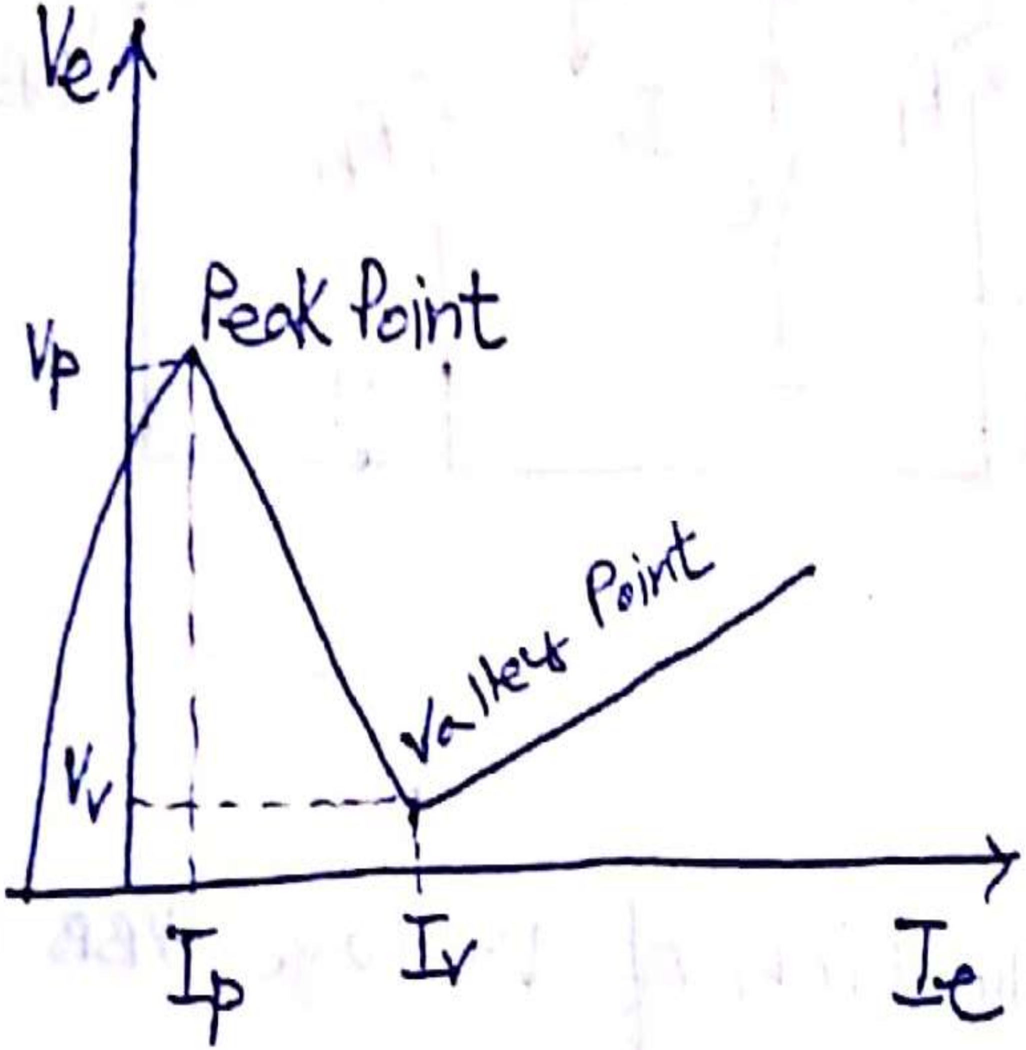
Potential of point A on application of voltage V_{BB} is

$$V_{AB1} = \frac{R_{B1}}{R_{B1}+R_{B2}} V_{BB} = \eta V_{BB}$$

η is termed the intrinsic stand off ratio.

$0.51 \leq \eta \leq 0.82$; $5\,K\Omega \leq R_{BB} \leq 10\,K\Omega$.

$R_{BB} = R_{B1} + R_{B2}$ & is known as interbase resistance.

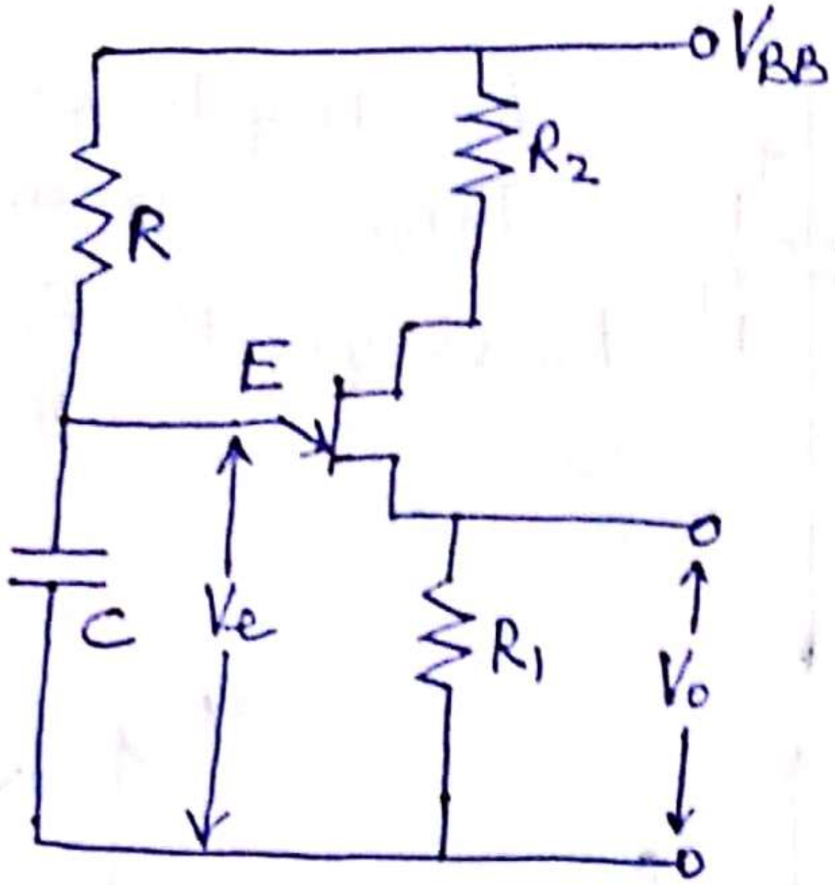


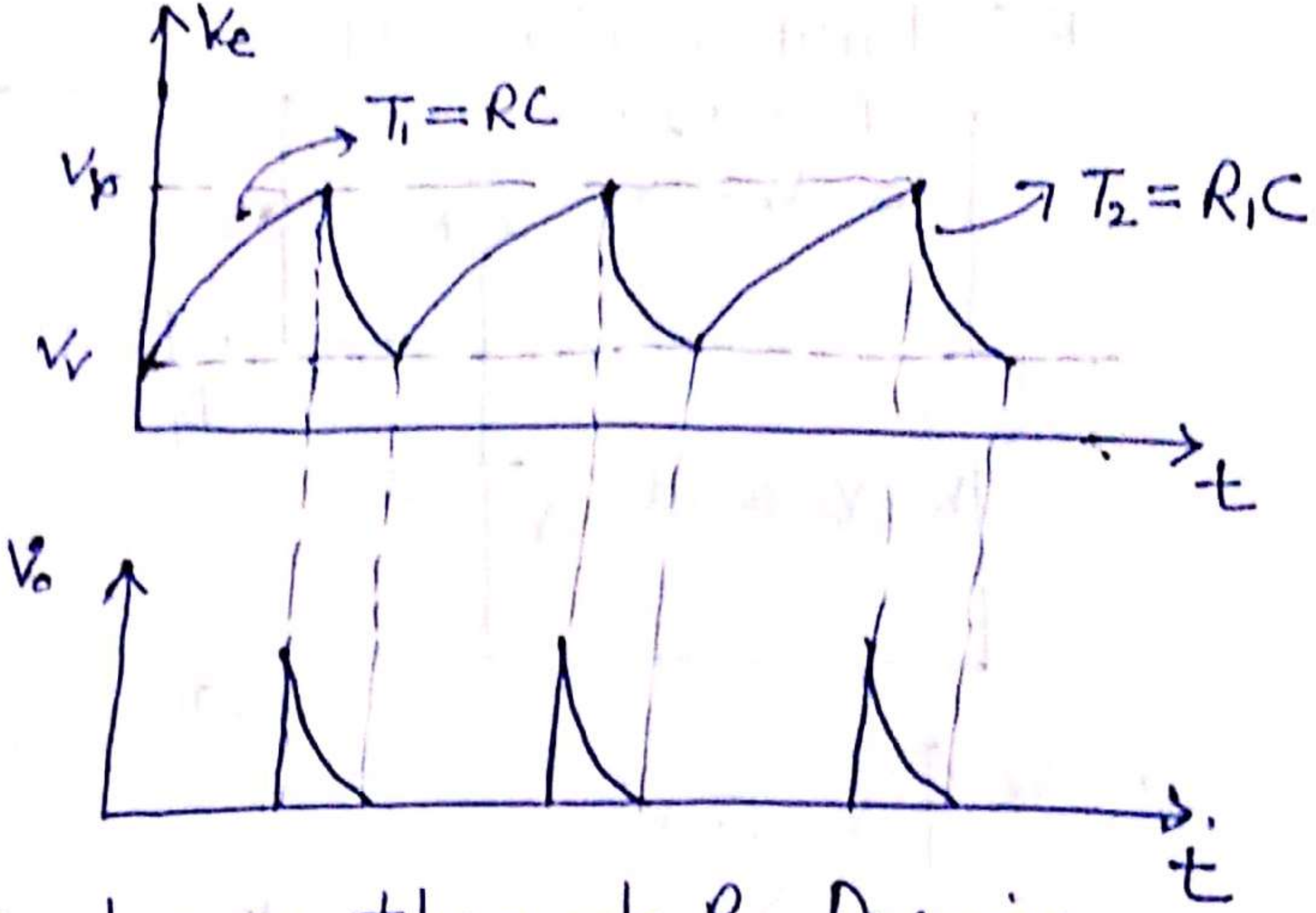
The negative resistance region between peak point & valley point gives UJT the switching characteristics for use in SCR triggering circuits.

∵ UJT exhibits -ve resistance characteristics, it can be used as a relaxation oscillator.

In electrical terminology, a relaxation oscillator is a circuit that repeatedly alternates between two states with a period that depends on the charging of a capacitor. The capacitor voltage may change exponentially when charged or discharged through a resistor from a constant voltage, or linearly when charged or discharged through a constant current source.

The period of oscillations is set by the time it takes for the system to relax from each disturbed state to the threshold that triggers the next disturbance.





When V_{BB} is applied, C charges through R. During charging, emitter circuit of UJT is open. Charging time constant $(\tau_1) = RC$.

When emitter voltage equals peak point voltage, unijunction between E & B_1 breaks down. UJT turns ON & C rapidly discharges through low resistance R_1, with a time constant $\tau_2 = R_1 C$. $\tau_2 \ll \tau_1$. The voltage drop across R_1, is applied to Gate-Cathode circuit of SCR, to turn it ON.

$T = RC \ln \frac{1}{1-\eta}$ where $T \rightarrow$ time required for capacitor to charge from V_v to V_p.